

Flight Routes Check

Time limit: 1.00 s

Memory limit: 512 MB

There are n cities and m flight connections. Your task is to check if you can travel from any city to any other city using the available flights.

Input

The first input line has two integers n and m : the number of cities and flights. The cities are numbered $1, 2, \dots, n$.

After this, there are m lines describing the flights. Each line has two integers a and b : there is a flight from city a to city b . All flights are one-way flights.

Output

Print "YES" if all routes are possible, and "NO" otherwise. In the latter case also print two cities a and b such that you cannot travel from city a to city b . If there are several possible solutions, you can print any of them.

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq m \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$

Example

Input	Output
4 5 1 2 2 3 3 1 1 4 3 4	NO 4 2